

Impact of the MW-M31 Merger on the Dark Matter Halo Distribution of M33

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ABSTRACT

Attempts to describe the dark matter halo distribution of a galaxy have resulted in three common geometrical prescriptions of the halo shape: prolate, oblate, and triaxial configurations. The precise form of the halo shape has been found to yield possible insight into the recent accretion and merger history of Milky Way-sized galaxies; however, this is less explored for satellite scale galaxies. In this work, I investigate the evolution of the M33 dark matter halo, the largest satellite in the Local Group, through collisionless N-body simulations of the Milky Way (MW), M31, and M33 system evolved 8 Gyr forward in time. The projected dark matter density distribution is geometrically analyzed near the time of the MW-M31 merger (~ 4 Gyr) to see if this event preferentially transforms the halo shape to prolate, oblate, or triaxial. I find that the pre-merger halo shape is loosely triaxial until the MW experiences a first pericenter approach with M31 at 4 Gyr, at which point the halo shape of M33 rapidly becomes more prolate, then turns more oblate as the MW recedes. This suggests that a prolate dark matter halo observed in a satellite galaxy may be associated with a recent event of tidal disruption. If satellite-sized prolate halos are unstable configurations, this could indicate that many subhalos in the early universe were prolate halos that did not survive long enough for a satellite to form that would be visible today.

Keywords: Local Group, Cold Dark Matter Theory, Dark Matter Halo, Halo Shape, Oblate/Prolate/Triaxial

1. INTRODUCTION

The majority of a galaxy’s mass is believed to come from its **dark matter halo**, a cloud-like structure of non-luminous matter that envelops the visible components of a galaxy. Existence of the dark matter halo was first hypothesized to explain the unusually high rotation speeds in the outskirts of nearby spiral galaxies, implying that there was significantly more matter present in these galaxies than could be observationally detected (F. Zwicky 1933; V. C. Rubin & W. K. Ford 1970). Although dark matter particles are invisible and weakly-interacting, the three dimensional distribution of the dark matter halo around a galaxy—known as the **halo shape**—is not considered to be random. Geometrical conventions such as **oblate** (elongated along poles), **prolate** (flattened at poles), or **triaxial** (asymmetric poles) matter distributions are commonly used metrics for quantifying the halo shape, based off of analyses of observational and simulated data.

Dark matter and baryons are the fundamental building blocks of a **galaxy**. Relating the dark matter halo shape to observable gas and stellar features in galax-

ies across cosmic time is essential for studying **galaxy evolution**, which correlates the morphology of a galaxy to a specific evolutionary stage. This paradigm forms the basis of the Lambda **cold dark matter** (Λ CDM) **theory** of cosmology, which asserts that dense regions of dark matter in the early universe provided the necessary scaffolding for galaxy formation by pulling in sufficient amounts of baryonic matter. Galaxies in the **Local Group**, namely the Milky Way (MW), M31, and M33, are ideal candidates to study dark matter halos because of their proximity, especially the halos of satellites that would otherwise be too distant to probe. This may lend valuable insight into a longstanding issue in Λ CDM cosmology: the missing satellites problem, which arises from the fact that fewer satellites have been found in the Local Group than predicted (G. Kauffmann et al. 1993).

Efforts to characterize the 3D shape of dark matter halos are largely driven by simulations, due to the invisible nature of dark matter. By examining Milky Way sized dark matter halos in the Aquarius cosmological N-body simulations, C. A. Vera-Ciro et al. (2011) found that halo shape measured at the virial radius typically

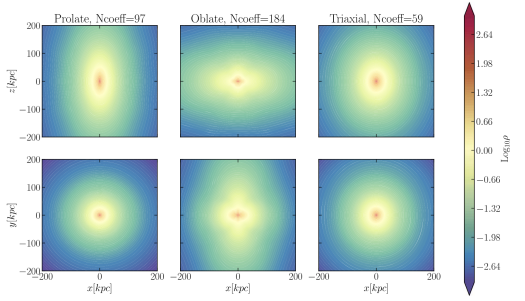


Figure 24. Density contours in the x - z and x - y planes of the prolate, oblate and triaxial halos are shown. The coefficients were computed up to $n_{\text{max}} = l_{\text{max}} = 20$ and the $\Gamma = 5$ was used to truncate the expansion. For the oblate halo the expansion needs high-order terms to reproduce the density field.

Figure 1. Figure 24 taken from N. Garavito-Camargo et al. (2021) displaying the morphology of prolate (left column), oblate (middle column), and triaxial (right column) dark matter halos. The top row is the x - z projection while the bottom row is x - y .

evolved from prolate to a more triaxial/oblate distribution. Narrow filaments of infalling material were found to correlate more with prolate configurations, while isotropic accretion gave rise to triaxial/oblate configurations. However, galaxy interactions may cause significant deviations from these three shapes. In N. Garavito-Camargo et al. (2021), N-body simulations of the LMC passing through the Milky Way showed that asymmetric perturbations were induced in the MW halo that did not fit a prolate, oblate, or triaxial distribution. Idealized versions of these three halo shapes computed by N. Garavito-Camargo et al. (2021) are shown in Figure 1 for reference.

The connection between dark matter subhalos and the missing satellites problem is an area of ongoing research. A thorough exploration into the geometrical evolution of satellite sized halos, as opposed to Milky Way sized halos, is not as heavily investigated, though simulations of galaxies in the Local Group have grown significantly in resolution. An interesting candidate is M33, as it is the third largest galaxy in the Local Group and lacks a central bulge common for spiral galaxies. Investigating how the shape of M33's dark matter halo evolves as it orbits M31 and the eventual merger between M31 and the MW could probe the stability of such satellite galaxies. A significant shift in halo configuration could be indicative of a process that might have destabilized many satellites in the early universe.

2. THIS PROJECT

In this paper, I aim to utilize the simulation data provided by R. P. van der Marel et al. (2012) of the MW-M31-M33 system to characterize the shape of M33's dark matter halo from the present day until ~ 12 Gyr in the future. I attempt to characterize the 3D halo shape

of M33 by examining isodensity contours of the dark matter halo distribution in different 2D projections.

This paper seeks to address how the halo shape is impacted by the MW-M31 merger in ~ 4 Gyr, if at all, and how this agrees with the predictions made by C. A. Vera-Ciro et al. (2011) about the evolution of halo shape but for satellite galaxies. If the distribution becomes more triaxial/oblate post-merger, then M33 may be experiencing tidal loss more isotropically. If the distribution instead becomes more prolate, then tidal disruption preferentially along the collision axis may be to blame. However, if the distribution remains the same, then a massive galaxy merger such as this may not have much of an effect on the halo shape of satellites.

Investigating how the halo shape of a M33-class satellite changes in response to a galaxy merger is important for understanding how dark matter halos evolved early in the universe when galaxy mergers were commonplace for large-scale structure formation. As a larger than average satellite, any significant changes to the halo shape of M33 can be generalized to lower mass satellites, helping to better understand satellite populations and their evolution.

3. METHODOLOGY

I rely on the data from R. P. van der Marel et al. (2012) produced using collisionless N-body simulations of the MW-M31-M33 system, which models the dynamics of particles belonging to the dark matter halo, disk, and bulge components of these galaxies. The gaseous component is not included in the simulations because of its substantially low fraction compared to the stellar mass in these galaxies. The dark matter halos are initially modeled with a Hernquist (1990) profile.

To evaluate the simulated halo distribution of M33, I use the high resolution dark matter particle data (particle type 1) to calculate the center of mass (COM) of the halo with the `CenterOfMass` code from Homework 4. The dark matter particles are used to calculate the COM as opposed to the disk particles because the halo COM will likely differ from the disk particle COM. The `RotateFrame` function from Lab 7 is applied to align the particle coordinates to create a reference frame where the x - y projection is a face-on view of M33, and the x - z and y - z projections are edge-on. The halo distribution can then be visualized in these frames by making a projected particle density plot using the `matplotlib` 2D histogram, shown in Figure 2. To extract useful information about the density distribution that can be related to the halo shape, I fit elliptical isodensity contours with `photutils` to achieve a similar plot to Figure 1. This fitting process measures the ellipticity of each of these

contours, shown in Figure 3. I streamline this entire process by creating my own code class called `HaloShape`.

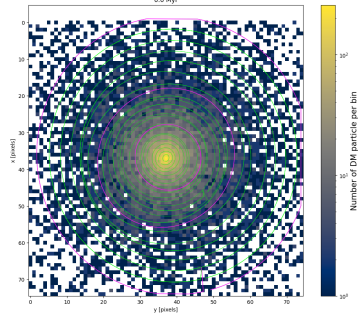


Figure 2. Face-on density projection (x-y) of the dark matter particles in M33 at Snapshot 0. Isodensity contours are plotted in green. The outer magenta line corresponds to roughly the halo R_{200} radius, followed by the $R_{200}/2$ and $R_{200}/4$ radii.

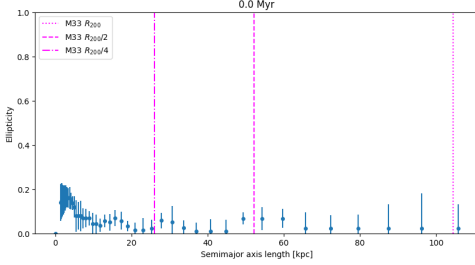


Figure 3. Corresponding ellipticity of the ellipses from Fig. 2 as a function of semimajor axis length in kpc. The dotted magenta line indicates the R_{200} radius, the dashed line being the $R_{200}/2$ radius, and the semi-dotted line being the $R_{200}/4$ radius.

The extent of the halo analyzed in this work is out to the R_{200} distance, which is considered the approximate edge of a dark matter halo. This distance is the radius at which the average spherical density of M33 is equal to 200 times the critical density of the universe, ρ_{crit} .

$$200\rho_{crit} = \frac{3M(< R_{200})}{4\pi R_{200}^3} \quad (1)$$

$M(< R_{200})$ is the halo mass of M33 within R_{200} . I base R_{200} off of the value for ρ_{crit} today using the Hubble constant defined by the cosmological parameters in Planck Collaboration et al. (2016), computed using the `CosmologicalTools` class from Lab 13. I assume ρ_{crit} to be constant throughout the simulation.

$$\rho_{crit} = \frac{3H_0^2}{8\pi G} \quad (2)$$

To find where R_{200} is in each snapshot, I calculate the average spherical density at a range of radii r , sampling

every 0.25 kpc from 0.25 kpc out to 200 kpc, and then identify at which radius the average density is closest to $200\rho_{crit}$. $M(< r)$ is computed by summing the masses of the DM particles within r using the `massEnclosed` method inside the `MassProfile` class. The maximum radial distance at which elliptical isodensity contours are fit is therefore R_{200} , in order to only consider the halo mass within the approximate virial radius of M33. To quantify the halo shape at R_{200} , I compute the triaxiality parameter T at this distance, defined by N. Garavito-Camargo et al. (2021) below:

$$T = \frac{1 - (b/a)^2}{1 - (c/a)^2} \quad (3)$$

where a , b , and c are the major, intermediate, and minor axes of the 3D halo shape ($a > b > c$). According to N. Garavito-Camargo et al. (2021), a prolate shape is observed for $T \geq 0.67$, oblate for $T \leq 0.33$, and triaxial for the range in between. Perfect triaxiality is achieved at $T = 0.5$. The values of b/a and c/a are both axis ratios (ARs) which can be calculated using the ellipticity f of the contour nearest to R_{200} via $AR = 1 - f$. Since $b/a > c/a$, I set b/a to be the axis ratio computed using the lowest ellipticity at R_{200} found across the x-y, y-z, and x-z projections. Similarly, the projection with the highest ellipticity value at R_{200} is used to compute the c/a axis ratio. Even though the true axis ratios may be different due to projection effects, they are a good enough ballpark estimate. To see if triaxiality varies within R_{200} , I also repeat this process at $R_{200}/2$ and $R_{200}/4$ to calculate corresponding triaxiality values.

The triaxiality evolution of the M33 halo shape can subsequently be analyzed by plotting the computed triaxiality values at every 10 snapshots. Using the definitions for prolate ($T \geq 0.67$), oblate ($T \leq 0.33$), and triaxial ($0.67 > T > 0.33$), the relationship between the halo shape of M33 and the dynamics of the M31-M33 (and eventually MW) system can be investigated, incorporating knowledge about their evolving orbital distances from Homework 6. The halo shape can also be visualized through any of the three density projections shortly before, during, and after the M31-MW merger has begun to see if any changes to the halo shape are observed.

I hypothesize that the halo distribution of M33 will be more triaxial at the present day because it has not been recently perturbed by M31 or any other object. As the M31-MW merger occurs, I predict that the halo shape of M33 will become more prolate due to the added potential of the Milky Way, elongating M33's halo along the collision axis due to tidal disruption.

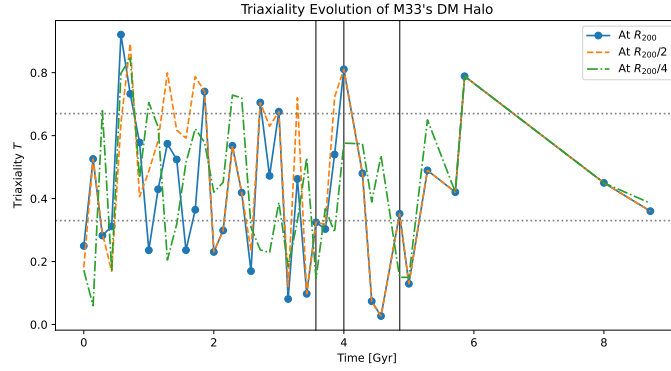


Figure 4. Computed triaxiality of the M33 DM halo at R_{200} , $R_{200}/2$, and $R_{200}/4$ as a function of time. The upper horizontal dotted line denotes $T = 0.67$ while the bottom one denotes $T = 0.33$. The vertical lines are added in reference to Figure 5. Dots are shown in the R_{200} line to indicate the sampling rate of snapshots, where each dot is a snapshot where the ellipse fitting was successful across all three projection axes.

4. RESULTS

The halo triaxiality evolution of M33 is plotted in Figure 4, from the present day to ~ 8 Gyr in the future. I was unable to fit ellipses to most halo distributions beyond 6 Gyr due to irregularities in the density distribution that caused the fitting algorithm to fail, creating gaps in snapshot sampling. The triaxiality at the R_{200} , $R_{200}/2$, and $R_{200}/4$ distances are plotted as separate lines. Horizontal lines indicating $T = 0.33$ and 0.67 are shown for reference. At the present day, the halo shape appears to be more oblate. Up until 4 Gyr, the shape at all three distances fluctuates within the triaxial regime, occasionally becoming more prolate or oblate in some places. Right before 4 Gyr and after, the triaxiality at these three distances generally follows the same trends: the halo is more oblate at 3.5 Gyr, after which it rapidly becomes more prolate by 4 Gyr, then drops down to being oblate again. The drastic changes in triaxiality at all three distances near the time of the MW-M31 merger at 4 Gyr suggest that the global halo shape is changing in response to the merger.

Figure 5 displays the evolution of the M33 DM halo from an face-on perspective (x-y projection) shortly before (left panel), at (middle panel), and after 4 Gyr (right panel). At 3.57 Gyr, the distribution looks somewhat elliptical, then is more elliptical at 4 Gyr and remains elliptical at 4.86 Gyr but with a more spread out distribution of matter. Vertical lines corresponding to these points in time are included in Figure 4. The left panel of Figure 5 therefore shows a more oblate shape before the merger and a more prolate shape around the 4 Gyr.

5. DISCUSSION

My analysis suggests that the DM halo shape of M33 at the present day is between being oblate and triaxial,

then momentarily becomes more prolate around the beginning of the MW-M31 merger, after which it becomes oblate again. I hypothesized that the halo shape would be more triaxial at the present day, so this result somewhat agrees with my prediction. The halo shape becoming more prolate in response to the MW-M31 merger, however, does agree with my hypothesis.

Using the mechanisms proposed by C. A. Vera-Ciro et al. (2011) that give rise to a prolate, oblate, or triaxial halo shape as a baseline, the sudden switch to a prolate configuration around the time of the MW-M31 merger may be because mass is being tidally pulled from M33 along the direction in which the MW and M31 are approaching each other. The halo shape appears to fluctuate in the triaxial regime between 0 and 3.5 Gyr, which is long enough that this may be normal behavior for the M33 halo, especially since these fluctuations are not resumed after 4 Gyr. The transition from a prolate to oblate configuration after 4 Gyr suggests that this may be the result of the MW moving away again from M31, thus reducing the strength of the tidal forces that would generate a prolate distribution. Overall, the halo spends the least amount of time in the prolate range, which may suggest that possessing a prolate configuration for an extended period of time will lead to the halo becoming unstable.

I limited my sampling in time to every 10 snapshots for computational efficiency but was unfortunately not always able to generate ellipticity values due to failures in the ellipse fitting process. Higher resolution sampling is needed to see if the trends I find in triaxiality are preserved, particularly the sharp jumps in triaxiality. A limitation of my methodology is the fact that the halo shape, in practice, is never perfectly prolate, oblate, or triaxial, therefore more irregular features in the halo shape are not usually resolvable by the ellipse contours.

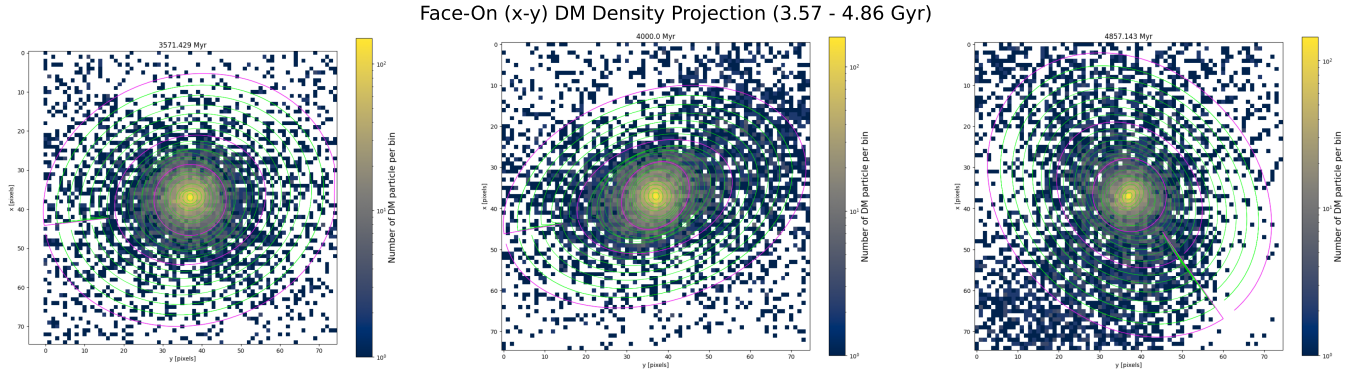


Figure 5. Density projections from a face-on perspective of the M33 DM halo at 3.57 Gyr (left panel), 4 Gyr (middle panel) and 4.86 Gyr (right panel). These snapshots were selected to show possible changes to the halo shape near the time of the MW-M31 merger.

Furthermore, as M33 gets closer to M31, R_{200} might not necessarily be a reasonable distance to evaluate the halo shape because the tidal forces will be much stronger. In these instances, the Jacobi radius should be considered to see if the mass at R_{200} is gravitationally bound to M33 or not. The projections I examine also do not necessarily coincide with the poles of the halo, thus it might not be possible at times to measure the maximum ellipticity because of these projection effects, which in turn affects the measured triaxiality.

6. CONCLUSIONS

Attempts to describe the dark matter halo distribution of a galaxy have resulted in three common geometrical prescriptions of the halo shape: prolate, oblate, and triaxial configurations. The precise form of the halo shape has been found to yield possible insight into the recent accretion and merger history of Milky Way-sized galaxies; however, this is less explored for satellite scale galaxies. In this work, I investigate the evolution of the M33 dark matter halo, the largest satellite in the Local Group, through collisionless N-body simulations of the Milky Way (MW), M31, and M33 system evolved 8 Gyr forward in time. The projected dark matter density distribution is geometrically analyzed near the time of the MW-M31 merger (4 Gyr) to see if this event preferentially transforms the halo shape to prolate, oblate, or triaxial.

I find that a rapid transition to a prolate halo shape in M33 is associated with the first pericenter approach of the MW relative to M31 around 4 Gyr. Once the orbital distance between the MW and M31 begins increasing again after first pericenter, the halo shape transitions to being more oblate, which is the same halo configuration achieved right before 4 Gyr. From the present

day to ~ 3.5 Gyr, without the added presence of the MW, the halo shape hovers around being triaxial. These findings are consistent with my initial hypothesis. A prolate configuration for a satellite-sized halo thus is more likely to show evidence of a recent event of tidal disruption, which may ultimately cause the halo to become unstable at this scale. In the context of the missing satellites problem, this could possibly mean that a large fraction of the first dark matter subhalos were prolate shaped and were thus unable to reproduce the satellites we detect today because of an instable halo environment.

Comparisons between the Jacobi radius and R_{200} should be done in future work to see at what snapshots R_{200} can be used as a valid distance to assess the halo shape or where it exceeds the Jacobi radius. A gaseous component should additionally be incorporated into this simulation to make the baryonic physics as accurate as possible. The response of the M33 stellar components to changes in the halo shape should also be explored to quantify the stability of a prolate configuration versus oblate or triaxial. This would be well supplemented by an analysis of satellite-sized halos in the early universe through cosmological simulations.

7. ACKNOWLEDGEMENTS

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Software citation information aggregated using **The Software Citation Station** (T. Wagg & F. S. Broekgaarden 2024; T. Wagg et al. 2024).

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